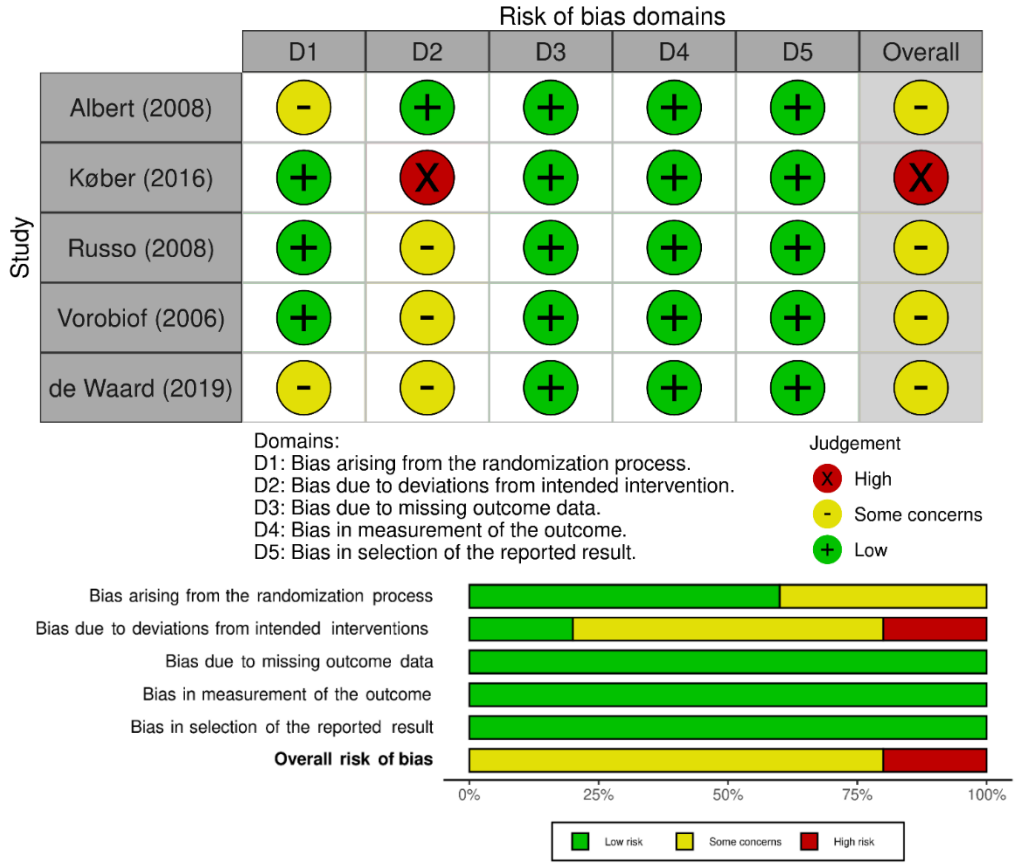
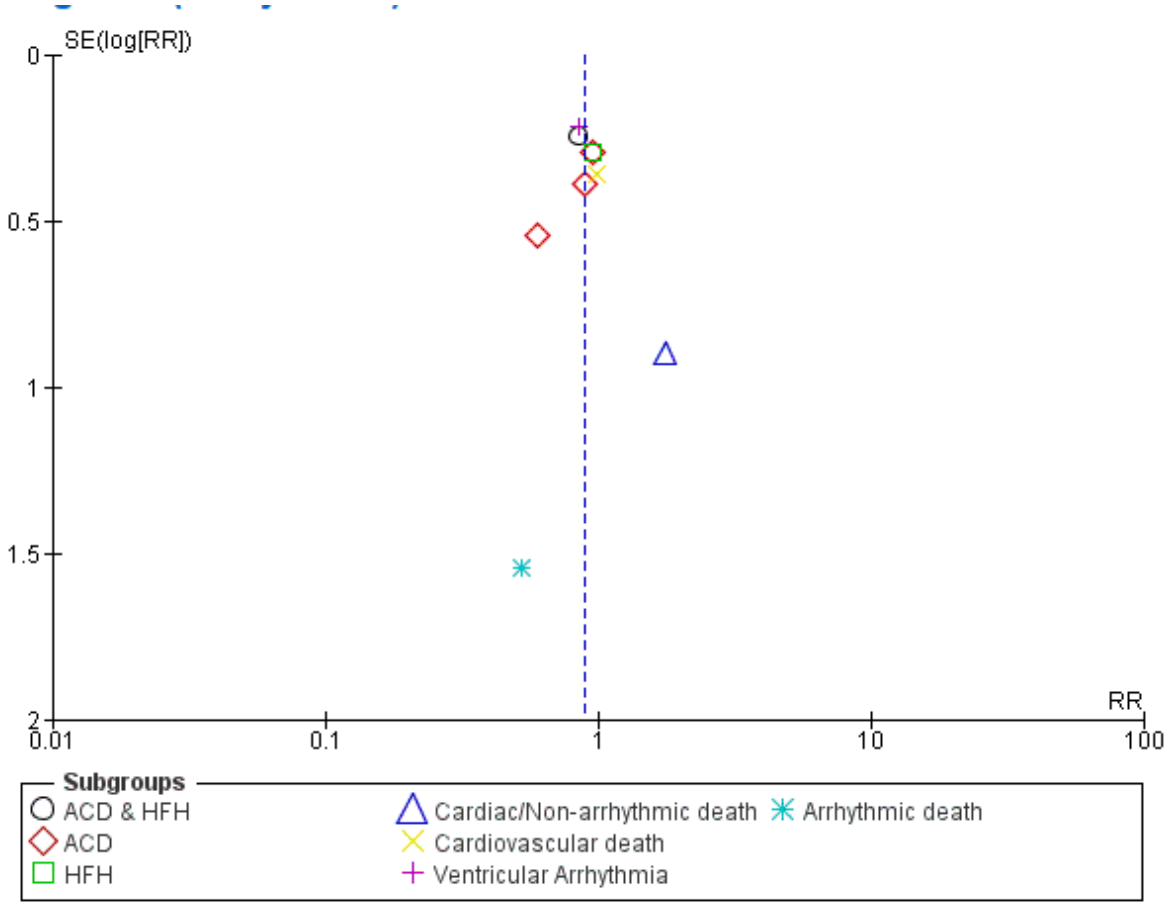
Supplementary Material

Supplemental Figure S1. Risk of Bias Analysis for Included ICD RCTs. Shown are the individual risk of bias domains for each study based on low (green), intermediate or some concerns (yellow) and high (red) study bias (top panel); and overall summary of each domain and the study analysis (bottom panel). **Abbreviations**: ICD = implantable cardioverter-defibrillator; RCT = randomized controlled trial.



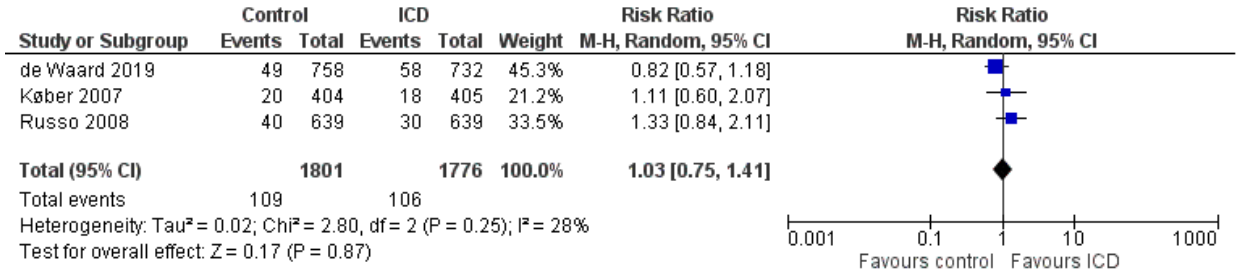
Supplemental Figure S2. Funnel Plots for Detection of Systematic Heterogeneity by Sex in ICD RCTs. Shown is a funnel plot for publication bias for patient cardiovascular (CV) outcomes by sex. Funnel plots pertaining to race and age were not included since a limited number of studies reported CV outcomes. Thus, the likelihood of discerning chance from real asymmetry was deemed too negligible to allow meaningful findings. **Abbreviations**: CV = cardiovascular; ICD = implantable cardioverter-defibrillator; RCT = randomized controlled trial.



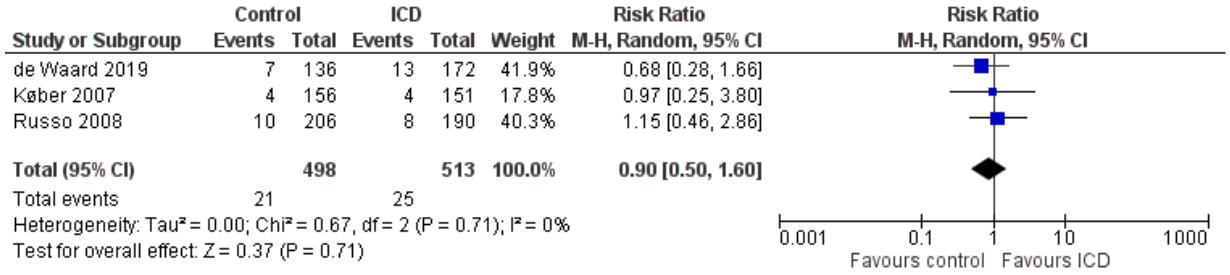
Supplemental Figure S3. Shown are Forest Plot analyses for all-cause death by sex between ICD treatment and control arms, with comparisons for males (**A**) and females (**B**). **Abbreviations:** ACD = all-cause death.

[a.] Only all-cause death was included in Forest plot analyses due to limited availability of data regarding cardiovascular outcomes.

A. Reported All-Cause Death in Males between ICD Treatment and Control Arms.



B. Reported All-Cause Death in Female between ICD Treatment and Control Arms.



Supplemental Table S1: Summary of Search Strategy for ICD RCTs

1	[Population: Cardiovascular Diseases]
2	exp Cardiovascular Diseases/ (2382985)
3	((cardiovascular or angiocardiovascular or angiocardiopath*) adj4 (disease* or disorder* or syndrome*)).tw,kw. (191327)
4	(stroke* or heart failure* or myocardial infarction or MI).tw,kw. (568645)
5	or/2-4 (2612784)
6	[Intervention: ICD]
7	Defibrillators, Implantable/ (16418)
8	(implant* adj3 (cardioverter* or defibrillator*)).tw,kw. (15607)
9	ICD.tw,kw. (35290)
10	or/7-9 (48879)
11	[Outcomes: MACE]
12	Death, Sudden, Cardiac/ (15442)
13	exp Mortality/ (382141)
14	exp Hospitalization/ and ((cardiac or heart) adj3 (failure or arrest)).tw,kw. (12633)
15	(outcome* or mortality or adverse event* or cardiovascular event* or death* or hospitali* or complication*).tw,kw. (3655691)
16	(MACE* or major adverse cardiovascular event* or major cardiovascular event*).tw,kw. (19020)
17	or/12-16 (3813340)
18	5 and 10 and 17 (12780)
19	[Study Types]
20	exp randomized controlled trial/ (511248)
21	exp Randomized Controlled Trials as Topic/ (138053)
22	(random* or placebo*).mp. (1447105)
23	(blind or double-blind).mp. (267067)
24	(clinical trial, all or controlled clinical trial or meta analysis or randomized controlled trial).pt,ti. (760278)
25	or/20-24 (1632329)
26	18 and 25 (2093)
27	limit 26 to "humans only (removes records about animals)" (2087)

Note: Sample search strategy is shown above for OVID (Medline) from 1946 to July 28, 2020. Similar search strategies were carried out for the other electronic databases. **Abbreviations**: ICD = implantable cardioverter-defibrillator; RCT = randomized controlled trial.

Supplemental Table S2. Summary of ICD RCTs excluded during data extraction

Study Characteristic	Jackson (2018)[1]	Russo (2009)[2]	Russo (2004)[3]	Jukema (2019)[4]	Hohnloser (2004)[5]	Tompkins (2015)[6]
Trial name	MADIT-RIT	INTRINSIC RV	MUSTT	ICD2	DINAMIT	MADIT- CRT
Trial sites region	North America, Europe, Asia	North America	North America	Europe	12 countries worldwide	North America, Europe
No. of trial sites	98	108	85	17	73	110
Eligible age	>21 years	>18 years	NR	≥55 years and <81 years	18 to 80 years	>21 years
Mean age	63 ± 11	63 ± 12	68	67 (median) (IQR: 62-74)	61.8	63 ± 11
Enrollment by sex (male vs female) (%)	1,064 (71%), 436 (29%)	1,236 (80.8%), 293 (19.1%)	71 (10%)	143 (76.1%), 45 (23.9%)	514 (76.3%), 160 (23.7%)	1,367 (75.1%), 453 (24.9%)
Male enrollment by race (White, Black, , Other) (%)[a]	829 (84.4%) White, 154 (15.6%) Black	NR	566 (89%) White, 70 (11%) Other.	NR	NR	NR
Female enrollment by race (White, Black, Asian, Other) (%)[a]	290 (71.1%) White, 118 (71%) Black	NR	56 (82%) White, 12 (18%) Other	NR	NR	NR
Patient population	Ischemic or non-ischemic heart disease, sinus rhythm, and met guidelines for primary prevention with	Standard indication for and ICD an ICD implanted, RV paced <20%	CAD, LVEF <40%, Sustained VT that was inducible with single, double, or triple extra-stimuli	Dialysis Patients with LVEF ≥35%	Recent MI (6 to 40 days previously) and a LVEF of 0.35 or less	Ischemic CM (NYHA class I or II) or non-ischemic CM (NYHA class II), sinus rhythm, EF <30%, prolonged QRS >130 msec

	an ICD or CRT-D					
Trial subgroup	MI, DCM	CAD, MI, DCM, ventricular arrhythmia	CAD, SCD, ventricular arrhythmia	SCD	MI	CM, CHF
Primary outcome(s) [b]	First occurrence of inappropriate therapy (therapy for non-ventricular tachyarrhythmia) *	HFH or death from any cause	Cardiac arrest or death from arrhythmia*	Sudden Cardiac (Arrhythmic) Death	Death from any cause	Death from any cause or a nonfatal HFH event*
Secondary outcome(s)	Death from any cause, first syncope	NR	Death from any cause, death from cardiac cause, or death from sustained ventricular arrhythmia	All-cause death, device related complications	Death due to Cardiac Arrhythmia*	Recurring HFH
Treatment arm	Tx1:High-rate ICD*; Tx2: delayed therapy ICD*	DDDR	EPG therapy and ICD*	ICD	ICD	CRT-ICD*
Control arm	Conventional therapy	VVI	EPG and no ICD	Usual Care (No ICD)	Conventional Medical Therapy (No ICD)	ICD
No. of patients (treatment vs control)	1,500 (Tx1: 500*, Tx2: 486*, vs 514)	988 (502 vs 486)	704 (351* vs 353)	188 (97 vs 91)	674 (332* vs 342)	1820 (1089* vs 731)
Reason for Exclusion	Three different programming arms which could not be	Patients randomized to different program settings, which could not be	Unable to retrieve event rates for outcomes in those who received an ICD or	No outcome event data.	Data not amenable to extraction as event rates could	No outcome values reported and only represented for ICD cohort so a meaningful

	compiled as one intervention.	compiled as one intervention.	antiarrhythmic therapy.		not be calculated from reported outcomes.	comparison could not be made.

[a] 'Other' race was any race that was non-white. The number of participants for race-based data by sex may not correspond to the total participants as data for other races in trials were negligible.

[b] Comparisons are between intervention versus control groups. (*) indicates significance at $P < 0.05$.

Abbreviations: DINAMIT, Defibrillator in Acute Myocardial Infarction Trial; DCM, dilated cardiomyopathy; HF, heart failure; HFH, heart failure hospitalization; ICD, implantable cardioverter-defibrillator; ICD2, Implantable Cardioverter-Defibrillator in Dialysis Patients Trial; INTRINSIV RV, Inhibition of Unnecessary RV Pacing with AVSH in ICDs; MADIT-RIT, Multicenter Automatic Defibrillator Implantation Trial–Reduce Inappropriate Therapy LVEF, left ventricular ejection fraction; MI, myocardial infarction; MADIT-CRT, Multicenter Automatic Defibrillator Implantation Trial with Cardiac Resynchronization Therapy; MUSTT, Multicenter Unsustain Tachycardia Trial; RCT, randomized controlled trial; SCD, sudden cardiac death; Tx, treatment; NR, not reported; DDDR, dual-chamber rate-responsive programming; VVI, single chamber programming; EPG, electrophysiological guided antiarrhythmic therapy.

Supplemental References:

1. Jackson LR, Thomas KL, Polonsky B, et al. Effectiveness of high rate and delayed detection ICD programming by race: A MADIT-RIT substudy. *Journal of cardiovascular electrophysiology* 2018;29(10):1418-24.
2. Russo AM, Day JD, Stolen K, et al. Implantable cardioverter defibrillators: do women fare worse than men? Gender comparison in the INTRINSIC RV trial. *Journal of cardiovascular electrophysiology* 2009;20(9):973-78.
3. Russo AM, Stamato NJ, Lehmann MH, et al. Influence of gender on arrhythmia characteristics and outcome in the Multicenter UnSustained Tachycardia Trial. *Journal of cardiovascular electrophysiology* 2004;15(9):993-98.
4. Jukema JW, Timal RJ, Rotmans JI, et al. Prophylactic use of implantable cardioverter-defibrillators in the prevention of sudden cardiac death in dialysis patients: the prospective, randomized, controlled ICD2 Trial. *Circulation* 2019;139(23):2628-38.
5. Hohnloser SH, Kuck KH, Dorian P, et al. Prophylactic use of an implantable cardioverter–defibrillator after acute myocardial infarction. *New England Journal of Medicine* 2004;351(24):2481-88.
6. Tompkins CM, Kutyifa V, Arshad A, et al. Sex differences in device therapies for ventricular arrhythmias or death in the Multicenter Automatic Defibrillator Implantation Trial with Cardiac Resynchronization Therapy (MADIT-CRT) trial. *Journal of cardiovascular electrophysiology* 2015;26(8):862-71.